Dictionary Expansion for Incompletely Overlapped Multi- and Hyperspectral Image Fusion

Xiaolin Han, *Member, IEEE*, Jiaxun Li, Wei Wang, *Member, IEEE*, Lijuan Niu, and Weidong Sun, *Member, IEEE*

***Abstract*— High-spatial-resolution multispectral (HM) and low-spatial-resolution hyperspectral (LH) image fusion over the same scene has been intensively studied. However, in practical application scenarios, the HM image usually covers a larger area than the LH image; thus, the same scene-oriented fusion framework can only be used to a limited range of the overlapping area. To solve this problem, a dictionary expansion-based incompletely overlapped HM and LH fusion method is proposed here, which tries to expand the dictionary learned in the overlapping area to the other nonoverlapping area and then to reconstruct an entire high-spatial-resolution hyperspectral (HH) image over the area covered by the whole HM image. Specifically, the above incomplete fusion problem is divided into one fusion problem in the overlapping area and one reconstruction problem over the nonoverlapping area separately, where the former is formulated as a traditional spectral dictionary learning-based fusion process and the latter is formulated as a new dictionary expansion-based reconstruction process only using the HM image in the framework of sparse and low-rank representation. To ensure the dictionary learned in overlapping areas could be precisely expanded to nonoverlapping areas, a strategy for extracting universal spectra is proposed by using the spectral similarity between overlapping and nonoverlapping areas. The expanded spectral dictionary is calculated using the spectral information from the LH image and the co-constraints applied to the representation error in both overlapping and nonoverlapping areas, and the corresponding sparse coefficient matrix is calculated using the spatial information from the HM image and a spectral similarity constraint between them. Experimental results on simulated and real datasets with different coverage areas indicate that our proposed method gives better performance in fusing incompletely overlapped multispectral and hyperspectral images, compared with other related or tangentially related state-of-the-art methods.**

***Index Terms*— Dictionary expansion, incompletely overlapped image fusion, multispectral and hyperspectral images.**

Received 21 October 2024; revised 26 March 2025 and 19 May 2025; accepted 29 June 2025. Date of publication 7 July 2025; date of current version 15 July 2025. This work was supported in part by the National Natural Science Foundation under Grant 41971294 and Grant 82471999, in part by Beijing Natural Science Foundation under Grant 3254044, and in part by Beijing Institute of Technology Research Fund Program for Young Scholars. *(Corresponding authors: Xiaolin Han; Weidong Sun.)*

Xiaolin Han and Jiaxun Li are with the School of Mechatronical Engineering, Beijing Institute of Technology, Beijing 100081, China (e-mail: hxl@bit.edu.cn; ljx@bit.edu.cn).

Wei Wang is with the School of Aerospace Engineering, Beijing Institute of Technology, Beijing 100081, China (e-mail: wangweiyh@bit.edu.cn).

Lijuan Niu is with the Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing 100021, China (e-mail: niulijuan8197@126.com).

Weidong Sun is with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: wdsun@tsinghua.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3586641

I. Introduction

HIGH-SPATIAL-RESOLUTION hyperspectral (HH) images with hundreds of narrow bands are more sensitive in identification [1], [2], segmentation [3], [4] of the surface features, and other applications [5], [6], [7]. However, as high spatial and high spectral resolution are mutually restricted with each other [8], [9], low-spatial-resolution hyperspectral (LH) images with limited area coverage and high-spatial-resolution multispectral (HM) images with limited bands are usually acquired simultaneously over the study area, how to reconstruct HH images by fusing the spectral information of LH images and the spatial information of HM images has attracted great attention.

Except for the extended pansharpening methods for each band of the HM image [10], [11], [12], current LH and HM image fusion methods can be roughly divided into dictionary-based, tensor-based, and deep learning-based methods. Focusing on improving the spatial resolution of LH images or improving the spectral resolution of HM images, the first category can be further subdivided into spatial- and spectral-dictionary-based methods.

Specifically, leveraging spatial dictionary pairs with high and low spatial resolution, Liang et al. [13] introduced an approach to enhance the spatial resolution of LH images. Utilizing local adaptive spatial dictionaries learned from neighboring patches, Zhao et al. [14] introduced a fusion approach based on collaborative representation. Arias et al. [15] introduced a nonlocal centralized sparse representation-based same scene-oriented fusion method, which constructs a spatial dictionary for each cluster. Besides, Sui et al. [16] suggested a Bayesian nonparametric-based fusion approach that learns the spatial dictionaries in subspace under sparse constraint. Based on Bayesian sparse representation, a fusion method was also introduced by Ghasrodashti et al. [17], which learns spatial dictionaries from the HM and an additional panchromatic image.

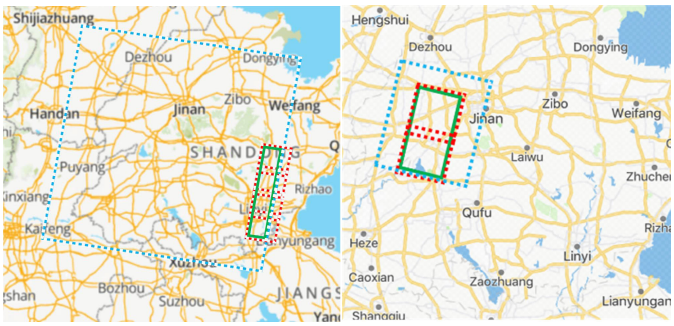
To deal with the spectral distortion problem in the above methods, a nonnegative matrix factorization-based fusion approach was offered by Yokoya et al. [18], where the endmembers can be regarded as a special kind of spectral dictionary. Regarding pure spectral signatures as a spectral dictionary, under sparse representation, Akhtar et al. [19] suggested a nonnegative-based approach to achieve the same scene LH and HM image fusion. With nonnegative constraints,

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

Han et al. [20] offered a sparse representation-based fusion approach, which takes the mixed spectral signatures as a spectral dictionary. Leveraging a spectral dictionary learned from the same scene LH image, Dong et al. [21] offered a nonlocal similarity-based sparse representation fusion approach. Han et al. [22] proposed a sparse representation and dictionary learning-based fusion method, which applies the spectral dictionary learned from the LH image directly to the HH image reconstruction. Besides, under the sparsity constraint of the residual term, a fusion approach introduced by Fu et al. [23] established a spectral dictionary by using the subspace learning methods. Utilizing superpixel-guided structure sparsity, the coupled spectral dictionaries were constructed by Zhang and Zhang [24], which are learned from the LH and the corresponding same-resolution multispectral image for LH and HM image fusion. Combined with weighted nuclear norm, Zhang et al. [25] suggested a fusion approach which learns the spectral dictionary from the LH image. To overcome the effect of misregistration and nonoverlapped bands, Zhang et al. [26] suggested a fusion approach, where the spectral dictionary consists of overlapping and nonoverlapping parts. Additionally, to simultaneously reconstruct the spectral and spatial information, a twin dictionary-based fusion method was offered by Han et al. [27], where the twin spectral–spatial dictionaries can represent the remaining spatial information and the spectral information, respectively. Xue et al. [28] offered a structured representation-based fusion approach, where the spectral dictionary and spatial dictionary are learned from the LH and HM images, respectively. Utilizing spatial–spectral features correlation, Yi et al. [29] introduced a spectral and spatial dictionary-based approach to fuse LH and HM images.

Regarding the hyperspectral image as a third-order tensor, tensor factorizations have been extensively used for LH and HM image fusion. Based on canonical polyadic (CP) decomposition, a coupled tensor factorization fusion method was supplied by Kanatsoulis et al. [30]. Considering unbalanced low-rankness of hyperspectral image in different modes, a Bayesian nonlocal CP factorization is suggested by Zeng et al. [31] to further enhance the fusion results, which addresses the rank selection problem. Based on Tucker decomposition, the same scene-oriented fusion approach to enhance the spatial resolution of LH images was developed by Dian et al. [32]. Combining the advantages of CP and Tucker factorization, a coupled tensor ring-based fusion method with low-rank constraints in the spectral domain was introduced by He et al. [33]. Besides, a tensor ring decomposition-based fusion method with weighted total variation and logarithmic nuclear norm for tensor ring factors was developed by Zhang et al. [34]. Utilizing higher-order factors, Xu et al. [35] proposed a tensor bifactor factorization method, which can effectively fuse same-scene LH and HM images.

Recently, utilizing a large number of training images a priori, deep learning-based fusion approaches have been widely studied and achieved better fusion results. Specifically, a fusion approach based on a pyramid fully convolutional neural network (CNN) was employed by Zhou et al. [36] to effectively utilize the spatial information of HM images.



(a) (b)

Fig. 1. Location and coverage of the typical HM and LH image pairs captured by the two onboard sensors of (a) HJ-1A and (b) CBERS-1D satellites.

Zhang et al. [37] offered a CNN-based approach to estimate the spatial and spectral degradations when fusing the same-scene LH and HM images. Peng et al. [38] proposed a CNN-based fusion method to improve the spectral resolution of multispectral images, which applies the nonlinear mapping established in the overlapping area to the nonoverlapping area directly, to predict the corresponding HH image. Considering the limitations of CNN-based methods in extracting global features, Hu et al. [39] proposed a visual Transformer-based fusion approach to learn spatial residual information. On this basis, fusion methods based on multiscale deep cross-Transformer [40], multiscale inception mixer Transformer [41], and double cross-Transformer [42] were subsequently employed to fully integrate spectral and spatial features. To overcome the difficulty of obtaining a large number of training images including the LH, HM, and HH images, an unsupervised network for LH and HM image fusion was suggested by Li et al. [43], which optimizes the network only by the observed LH–HM image pair. Utilizing backpropagation neural network (BPNN), a multibranch-based fusion approach was suggested by Han et al. [44], which learns the spectral mappings for each cluster at low spatial resolution and applies it to high spatial resolution, to reconstruct the HH image. Besides, Liu et al. [45] introduced an unsupervised fusion method based on a lightweight-attention cyclic network, which requires few parameters to extract multiscale features. Yu et al. [46] suggested an unsupervised spatial reconstruction network for image fusion, which learns the spectral mapping and the differences between heterogeneous images.

In most of the existing multispectral and hyperspectral image fusion methods, at least a pair of images, including an HM and an LH image over the same scene, is needed. However, in practical application scenarios, the HM image usually covers a larger area than the LH image, at which time, the HH image in the nonoverlapping area is challenging to obtain as it does not satisfy the existing fusion model and that limits the coverage of the fused HH image and the applicability for subsequent applications. Taking the typical Huan Jing Satellite-1A of China (HJ-1A) [47] and China-Brazil Earth Resource Satellite-1D (CBERS-1D) [48] as an example, Fig. 1 gives the location and coverage of the HM and LH image pairs captured by the two onboard sensors of HJ-1A and CBERS-1D satellites over Shandong Province of China. Among them,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

the spatial resolution of the HM and LH images of the HJ-1A satellite is 30 and 100 m, and those of the CBERS-1D satellite are 10 and 30 m, respectively. As can be seen from Fig. 1, the blue and red dashed boxes indicate the HM and LH images, respectively, where the green boxes denote the overlapping areas covered by both the HM and LH images, while most of remaining parts of HM images in the blue dashed boxes indicate the nonoverlapping areas without the corresponding LH image. This phenomenon indicates that the incompletely overlapped fusion problem can commonly be found in the synchronous observation by two different HM and LH sensors onboard the same satellite. Strictly speaking, due to the observation width of the LH image being usually narrower than that of the HM image, the incompletely overlapped fusion problem exists in most of the fusion processes for the entire area. On the one hand, most of the existing fusion methods usually take a strategy that, only crops an accurately corresponded HM and LH image pair on the overlapping area for the fusion, but discards the remaining nonoverlapping areas which may be relatively larger than the overlapping area, to avoid the problem of incomplete coverage between them. On the other hand, so far, few methods specifically address this problem, only several tangentially related methods, such as spectral dictionary-based method [22] and spectral mapping-based methods [38], [43], [44], [45], [46], can be applied to reconstruct the entire HH images to some extent, by directly applying the spectral dictionary or the spectral mapping relationship established in the overlapping area to the nonoverlapping area. Although using the above methods can obtain the entire HH image over the full observation area, the relationship fixed by the pretrained neural network model or dictionary is hard to fine-tune, due to the lack of corresponding samples in the nonoverlapping area. Meanwhile, the direct application of this kind of fixed model or dictionary cannot address the differences between the overlapping and nonoverlapping areas. That is to say, although there are many similar spectra between the overlapping and nonoverlapping areas that can be used to fine-tune or optimize the pretrained model or dictionary between HM and LH images for the nonoverlapping areas, it is extremely challenging to determine in what form and how to embed these kinds of new information into the pretrained model or dictionary, to achieve a more accurate fusion performance for the incompletely overlapping area.

To address the above problems, a dictionary expansion-based incompletely overlapped HM and LH fusion method (termed as DEF) is proposed in this article, try to expand the spectral dictionary learned within the overlapping area to the other nonoverlapping area and to reconstruct an entire HH image over the area covered by the whole HM image. Specifically, first, a dictionary expansion scheme is established, which models the incomplete image fusion problem as one fusion problem in the overlapping area and one reconstruction problem over the nonoverlapping area, separately. Second, the fusion problem in the overlapping area is formulated as a traditional spectral dictionary learning based fusion process. Third, to ensure the dictionary learned in overlapping areas could be precisely expanded to nonoverlapping areas, a strategy for extracting universal spectra is proposed by using the spectral similarity between overlapping and nonoverlapping areas. Then, based on the expanded spectral dictionary and the universal spectra, the reconstruction problem over the nonoverlapping area is formulated as a dictionary expansion-based reconstruction process only using the HM image under sparse and low-rank constraints. In which the expanded spectral dictionary is optimized using the spectral information from the LH image and a representation error co-constraint in both overlapping and nonoverlapping areas, the corresponding sparse coefficient matrix is optimized using the spatial information from the HM image and a spectral similarity constraint between them. Finally, the HH image is obtained by merging the fused image in the overlapping area and the reconstructed image over the nonoverlapping area. The major contributions are listed as follows.

1) A new dictionary expansion scheme for incompletely overlapped HM and LH image fusion is proposed for the first time, utilizing the spectral information from the overlapping area and the spatial information from the entire area.
2) A strategy to expand the spectral dictionary learned in overlapping areas to nonoverlapping areas with the help of universal spectra between them is proposed, and its optimization is ensured by the co-constraints of overlapping and nonoverlapping areas.
3) For the reconstruction over the nonoverlapping area, a closed-form solution for the optimized sparse coefficient matrix is calculated theoretically, using the spatial information from the nonoverlapped HM image with a spectral similarity constraint.

The remainder of this article is presented as follows. The proposed DEF method is introduced in Section II. Section III gives experimental results on the simulated and real datasets, followed by the conclusion in Section IV.

II. Proposed Method

A. Problem Formulation

In the incompletely overlapped HM and LH fusion, the HH image $\mathbf{X} \in \mathbb{R}^{\lambda_x \times N}$ covering the entire area is expected to be reconstructed by fusing the HM image $\mathbf{Y}_m \in \mathbb{R}^{\lambda_y \times N}$ of the entire area and the LH image $\mathbf{Y}_h \in \mathbb{R}^{\lambda_x \times m}$ of the overlapping area only, where λ_x and λ_y represent the number of spectral bands in $\mathbf{X}$ and $\mathbf{Y}_m$, respectively, and N and m represent the number of pixels in each band of $\mathbf{X}$ and $\mathbf{Y}_h$, respectively. Among them, the HH image $\mathbf{X}$ is divided into two parts: $\mathbf{X}^s = \mathbf{XS}$ denotes the overlapped HH image and $\mathbf{X}^r = \mathbf{XS}^r$ denotes the remaining nonoverlapped HH image, where $\mathbf{S} \in \mathbb{R}^{N \times M}$ and $\mathbf{S}^r \in \mathbb{R}^{N \times (N-M)}$ $(M < N)$ denote the mask matrixes to express the overlapping area covered by $\mathbf{Y}_h$, and the remaining nonoverlapping area without the corresponding LH image, respectively. Moreover, $\mathbf{S}$ and $\mathbf{S}^r$ are sparse matrices with only one "1" in each column and the rest are "0." With this, the LH image $\mathbf{Y}_h$ and the HM image $\mathbf{Y}_m$ can be regarded as spatial and spectral degradations of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

overlapped HH image $\mathbf{X}^s$ and HH image $\mathbf{X}$

$$\mathbf{Y}_h = \mathbf{X}^s\mathbf{B} + \mathbf{N}_h \tag{1}$$

$$\mathbf{Y}_m = \mathbf{L}\mathbf{X} + \mathbf{N}_m \tag{2}$$

where $\mathbf{L} \in \mathbb{R}^{\lambda_y \times \lambda_x}$ represents the spectral response function, $\mathbf{B} \in \mathbb{R}^{M \times m}$ represents the blurring and downsampling process, and $\mathbf{N}_h \in \mathbb{R}^{\lambda_x \times m}$ and $\mathbf{N}_m \in \mathbb{R}^{\lambda_y \times N}$ denote additive Gaussian noises [21], [27].

Utilizing sparse representation in the spectral domain to solve the above fusion problem, the overlapped HH image $\mathbf{X}^s$ can be expressed by the spectral dictionary $\mathbf{D}_s \in \mathbb{R}^{\lambda_x \times K}$

$$\mathbf{X}^s = \mathbf{D}_s\mathbf{A}^s + \mathbf{N}^s \tag{3}$$

where $\mathbf{A}^s \in \mathbb{R}^{K \times M}$ and $\mathbf{N}^s \in \mathbb{R}^{\lambda_x \times M}$ denote the sparse coefficient matrix and the additive Gaussian noise, respectively. Considering that $\mathbf{X}^s$ and $\mathbf{X}^r$ are neighboring areas with a high degree of ground surface similarity, the spectral dictionary $\mathbf{D}_s$ used to represent the overlapped HH image $\mathbf{X}^s$ can be expanded as an initial value to represent the nonoverlapped HH image $\mathbf{X}^r$, which is optimized in Section II-C

$$\mathbf{X}^r = \mathbf{D}_s\mathbf{A}^r + \mathbf{N}^r \tag{4}$$

where $\mathbf{A}^r \in \mathbb{R}^{K \times (N-M)}$ and $\mathbf{N}^r \in \mathbb{R}^{\lambda_x \times (N-M)}$ denote the corresponding sparse coefficient matrix and Gaussian noise, respectively. Substituting (3) and (4) into (1) and (2) leads to the following equations:

$$\mathbf{Y}_h = \mathbf{D}_s\mathbf{A}^s\mathbf{B} + \mathbf{N}_h^s \tag{5}$$

$$\mathbf{Y}_m^s = \mathbf{L}\mathbf{D}_s\mathbf{A}^s + \mathbf{N}_m^s \tag{6}$$

$$\mathbf{Y}_m^r = \mathbf{L}\mathbf{D}_s\mathbf{A}^r + \mathbf{N}_m^r \tag{7}$$

where the HM image $\mathbf{Y}_m$ is further divided into an overlapped HM image $\mathbf{Y}_m^s = \mathbf{Y}_m\mathbf{S}$ and a remaining nonoverlapped HM image $\mathbf{Y}_m^r = \mathbf{Y}_m\mathbf{S}^r$, and $\mathbf{N}_m^s \in \mathbb{R}^{\lambda_y \times M}$ and $\mathbf{N}_m^r \in \mathbb{R}^{\lambda_y \times (N-M)}$ denote the Gaussian noises. Taking the typical HJ-1A and CBERS-1D satellites as an example, the green boxes given in Fig. 1 indicate the overlapping areas with the overlapped HM image $\mathbf{Y}_m^s$ and LH image $\mathbf{Y}_h$, while the remaining part in the blue dashed box indicates the nonoverlapped HM image $\mathbf{Y}_m^r$.

As can be seen from (5) to (7), the incompletely overlapped multispectral and hyperspectral image fusion has been divided into two parts, one is to fuse the HH subimage $\mathbf{X}^s$ in the overlapped area by using the given LH image $\mathbf{Y}_h$ with the HM image $\mathbf{Y}_m^s$, and the other is to reconstruct the rest of the entire HH image $\mathbf{X}^r$ in the nonoverlapping area only from the remaining HM image without the corresponding LH image. The former can be solved via the optimization of (5) and (6) by the widely applied spectral dictionary based fusion methods [18], [21], [27]. While the latter has been transformed into an optimization of expanded spectral dictionary $\mathbf{D}_s$ and its sparse coefficient matrix $\mathbf{A}^r$.

B. Sparse Coefficient Optimization

As discussed above, to reconstruct the nonoverlapped HH image $\mathbf{X}^r$ using (5)–(7), spectral dictionary $\mathbf{D}_s$ and its sparse coefficient matrix $\mathbf{A}^r$ are the two major matrices that we should solve and optimize. And apart from sparse constraint, since $\mathbf{X}^s$ and $\mathbf{X}^r$ are neighboring areas, some universal spectra in $\mathbf{X}^s$, denoted as $\mathbf{X}_u^s = \mathbf{X}^s\mathbf{U} = \mathbf{D}_s\mathbf{A}_u^s + \mathbf{N}_u^s$, are more similar to the spectrum in $\mathbf{X}^r$, where $\mathbf{A}_u^s = \mathbf{A}^s\mathbf{U}$, $\mathbf{N}_u^s = \mathbf{N}^s\mathbf{U}$, $\mathbf{U} \in \mathbb{R}^{M \times n}$ $(n < M)$ denotes the universal matrix with only one "1" in each column and the rest are "0." Therefore, the combination of the corresponding sparse coefficient matrix $[\mathbf{A}^r\ \mathbf{A}_u^s]$ has low-rank characteristic under the same dictionary $\mathbf{D}_s$ representation. Then, the optimization becomes

$$\begin{aligned}\arg\min_{\mathbf{D}_s,\mathbf{A}^r}\ & \eta\left\|\mathbf{Y}_h - \mathbf{D}_s\mathbf{A}^s\mathbf{B}\right\|_F^2 + \left\|\mathbf{Y}_m^s - \mathbf{L}\mathbf{D}_s\mathbf{A}^s\right\|_F^2 \\ & + \left\|\mathbf{Y}_m^r - \mathbf{L}\mathbf{D}_s\mathbf{A}^r\right\|_F^2 + \lambda\left\|\mathbf{A}^r\right\|_1 + \gamma\left\|\left[\mathbf{A}^r\ \mathbf{A}_u^s\right]\right\|_*. \end{aligned} \tag{8}$$

Here, l_1 and nuclear norms are natural convex approximations of sparse and low-rank constraints, respectively [28], [34]. Moreover, the spectral similarity between $\mathbf{X}^r$ and universal spectra $\mathbf{X}_u^s$ also reflects in its spectral degradation domain, that is, the spectral similarity between $\mathbf{Y}_m^r$ and $\mathbf{Y}_u^s = \mathbf{Y}_m^s\mathbf{U}$, which can be measured by

$$\boldsymbol{u}(i) = \min_j \frac{\boldsymbol{y}_m^s(i)^{\mathbf{T}}\boldsymbol{y}_m^r(j)}{\left\|\boldsymbol{y}_m^s(i)\right\|_2\left\|\boldsymbol{y}_m^r(j)\right\|_2} \tag{9}$$

where $\boldsymbol{y}_m^s(i), \boldsymbol{y}_m^r(j) \in \mathbb{R}^{\lambda_y}$ denote the ith and jth spectra in $\mathbf{Y}_m^s$ and $\mathbf{Y}_m^r$, respectively, $\boldsymbol{y}_m^s(i)^{\mathbf{T}}$ denotes the transpose matrix of $\boldsymbol{y}_m^s(i)$, and the similarity index $\boldsymbol{u}(i)$ denotes the ith atom in $\boldsymbol{u} \in \mathbb{R}^M$. In terms of formula expression, different from the spectral angle mapper (SAM) index, which is usually used to evaluate a one-to-many relationship [49], the spectral similarity $\boldsymbol{y}_m^s(i)^{\mathbf{T}}\boldsymbol{y}_m^r(j)/||\boldsymbol{y}_m^s(i)||_2||\boldsymbol{y}_m^r(j)||_2$ measure in (9) is a many-to-many relationship between two big spectral sets $\mathbf{Y}_m^r$ and $\mathbf{Y}_u^s$, in which the arccosine calculation is removed to avoid mass computation. As can be seen from (9), it calculates the lower limit of spectral similarity and forms the similarity index matrix $\boldsymbol{u}$. And, the similarity index $\boldsymbol{u}(i)$ will be higher, when $\boldsymbol{y}_m^s(i)$ is more similar to the spectra in $\mathbf{Y}_m^r$. Accordingly, the universal matrix $\mathbf{U}$ achieves taking the corresponding spectra whose similarity is larger than the threshold σ as the universal spectra $\mathbf{X}_u^s$, described as

$$\begin{aligned}&\mathbf{U}(i, j) = \delta(i - \boldsymbol{u}_s(j)) \\ &s.t.\ \boldsymbol{u}_s = \{k|\boldsymbol{u}(k) > \sigma\}, \quad k = 1, \ldots, M\end{aligned} \tag{10}$$

where the threshold σ denotes the nth largest value in the similarity index $\boldsymbol{u}$, δ denotes the Dirac delta function, $\boldsymbol{u}_s \in \mathbb{R}^n$ denotes the index of top n value in $\boldsymbol{u}$, and the similarity ratio of n to M can be denoted as ξ. As can be seen from (9) and (10), it attempts to find n spectra with the highest similarity from a set of M elements and another set of $N - M$ elements via similarity index matrix $\boldsymbol{u}$, to obtain the corresponding universal spectra $\mathbf{X}_u^s$.

As shown in (8), there are two matrices we need to optimize simultaneously. Normally, this process of simultaneously optimizing two parameters can be decomposed into two subproblems, that is, solving one of them individually each time. Here, as the initial value of spectral dictionary $\mathbf{D}_s$ is already given, we can first fix $\mathbf{D}_s$ to optimize $\mathbf{A}^r$ and then fix $\mathbf{A}^r$ to optimize $\mathbf{D}_s$. With the fixed spectral dictionary $\mathbf{D}_s$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

the optimization of the sparse coefficient matrix $\mathbf{A}^r$ in (8) can also be rewritten as

$$\arg\min_{\mathbf{A}^r} \left\|\mathbf{Y}_m^r - \mathbf{L}\mathbf{D}_s\mathbf{A}^r\right\|_F^2 + \lambda\left\|\mathbf{A}^r\right\|_1 + \gamma\left\|\left[\mathbf{A}^r\ \mathbf{A}_u^s\right]\right\|_*. \quad (11)$$

To separately minimize each term, the ADMM algorithm [50] is used with splitting variables $\mathbf{E} = \mathbf{A}^r$ and $\mathbf{F} = [\mathbf{A}^r\ \mathbf{A}_u^s]$. Then, (11) becomes

$$\begin{aligned}&\arg\min_{\mathbf{A}^r} \left\|\mathbf{Y}_m^r - \mathbf{L}\mathbf{D}_s\mathbf{A}^r\right\|_F^2 + \lambda\|\mathbf{E}\|_1 + \gamma\|\mathbf{F}\|_* \\ &s.t.\ \mathbf{E} = \mathbf{A}^r, \mathbf{F} = \left[\mathbf{A}^r\ \mathbf{A}_u^s\right].\end{aligned} \quad (12)$$

To optimize each variable separately, the above problem can be optimized by

$$\begin{aligned}&\arg\min_{\mathbf{E}}\ \lambda\|\mathbf{E}\|_1 + \mu_1\left\|\mathbf{E} - \mathbf{A}^r + \frac{\mathbf{V}_1}{2\mu_1}\right\|_F^2 \\ &\arg\min_{\mathbf{F}}\ \gamma\|\mathbf{F}\|_* + \mu_1\left\|\mathbf{F} - \left[\mathbf{A}^r\ \mathbf{A}_u^s\right] + \frac{\mathbf{V}_2}{2\mu_1}\right\|_F^2 \\ &\arg\min_{\mathbf{A}^r}\ \left\|\mathbf{Y}_m^r - \mathbf{L}\mathbf{D}_s\mathbf{A}^r\right\|_F^2 + \mu_1\left\|\mathbf{E} - \mathbf{A}^r + \frac{\mathbf{V}_1}{2\mu_1}\right\|_F^2 \\ &\qquad + \mu_1\left\|\mathbf{F}\mathbf{H} - \mathbf{A}^r + \frac{\mathbf{V}_2\mathbf{H}}{2\mu_1}\right\|_F^2\end{aligned} \quad (13)$$

which is calculated by

$$\begin{aligned}&\mathbf{E} = \text{soft}\left(\mathbf{A}^r - \frac{\mathbf{V}_1}{2\mu_1}, \frac{\lambda}{2\mu_1}\right) \\ &\mathbf{F} = \mathbf{U}\text{soft}\left(\Sigma, \frac{\gamma}{2\mu_1}\right)\mathbf{V}^{\mathbf{T}} \\ &\mathbf{A}^r = \left(\mathbf{D}_l^{\mathbf{T}}\mathbf{D}_l + 2\mu_1\mathbf{I}\right)^{-1}\left[\mu_1\left(\mathbf{F}\mathbf{H} + \frac{\mathbf{V}_2\mathbf{H}}{2\mu_1}\right) + \right. \\ &\qquad \left. + \mathbf{D}_l^{\mathbf{T}}\mathbf{Y}_m^r + \mu_1\left(\mathbf{E} + \frac{\mathbf{V}_1}{2\mu_1}\right)\right].\end{aligned} \quad (14)$$

Among them, $\mathbf{H} = [\mathbf{I}_{(N-M)\times(N-M)}\ \mathbf{0}_{(N-M)\times n}]^{\mathbf{T}}$, $\mathbf{D}_l = \mathbf{L}\mathbf{D}_s$, the singular value decomposition [51] of $[\mathbf{A}^r\ \mathbf{A}_u^s] - (\mathbf{V}_2/2\mu_1)$ obtains $\mathbf{U} \in \mathbb{R}^{K\times K}$, $\Sigma \in \mathbb{R}^{K\times(N-M+n)}$ and $\mathbf{V} \in \mathbb{R}^{(N-M+n)\times(N-M+n)}$, $\mathbf{V}_1 \in \mathbb{R}^{K\times(N-M)}$ and $\mathbf{V}_2 \in \mathbb{R}^{K\times(N-M+n)}$ denote Lagrangian multipliers, and soft(u, v) denotes the soft-threshold function [27], [34].

C. Expanded Spectral Dictionary Optimization

After optimizing the sparse coefficient matrix $\mathbf{A}^r$ via (14) and estimating $\mathbf{A}^s$ by the existing same scene-oriented fusion methods [18], [21], [27], the optimization of spectral dictionary $\mathbf{D}_s$ can be rewritten as

$$\begin{aligned}\arg\min_{\mathbf{D}_s}\ &\eta\left\|\mathbf{Y}_h - \mathbf{D}_s\mathbf{A}_b^s\right\|_F^2 + \left\|\mathbf{Y}_m^s - \mathbf{L}\mathbf{D}_s\mathbf{A}^s\right\|_F^2 \\ &+ \left\|\mathbf{Y}_m^r - \mathbf{L}\mathbf{D}_s\mathbf{A}^r\right\|_F^2\end{aligned} \quad (15)$$

where $\mathbf{A}_b^s = \mathbf{A}^s\mathbf{B}$. In (15), the first and second terms constrain the representation error in the overlapping area, while the last term constrains that in the nonoverlapping area. The above co-constraints applied to the representation error in both overlapping and nonoverlapping areas suggest that, spectral dictionary $\mathbf{D}_s$ can be optimized and expanded from the overlapping area to the nonoverlapping area by using the spectral information of overlapped LH image via (15). So, in this sense, spectral dictionary $\mathbf{D}_s$ can also be regarded as an expanded spectral dictionary. Utilizing a Lagrangian multiplier $\mathbf{V}_3 \in \mathbb{R}^{\lambda_y\times N}$ and the splitting variable $\mathbf{G} = \mathbf{D}_s[\mathbf{A}^s\ \mathbf{A}^r]$, (15) can be reformulated into

$$\begin{aligned}\arg\min_{\mathbf{D}_s,\mathbf{G}}\ &\eta\left\|\mathbf{Y}_h - \mathbf{D}_s\mathbf{A}_b^s\right\|_F^2 + \left\|\left[\mathbf{Y}_m^s\ \mathbf{Y}_m^r\right] - \mathbf{L}\mathbf{G}\right\|_F^2 \\ &+ \mu_2\left\|\mathbf{G} - \mathbf{D}_s\left[\mathbf{A}^s\ \mathbf{A}^r\right] + \frac{\mathbf{V}_3}{2\mu_2}\right\|_F^2.\end{aligned} \quad (16)$$

With $\mathbf{A} = [\mathbf{A}^s\ \mathbf{A}^r]$, spectral dictionary $\mathbf{D}_s$ can be calculated via the following closed-form solutions:

$$\begin{aligned}&\mathbf{G} = \left(\mathbf{L}^{\mathbf{T}}\mathbf{L} + \mu_2\mathbf{I}\right)^{-1}\left[\mathbf{L}^{\mathbf{T}}\left[\mathbf{Y}_m^s\ \mathbf{Y}_m^r\right] + \mu_2\left(\mathbf{D}_s\mathbf{A} - \frac{\mathbf{V}_3}{2\mu_2}\right)\right] \\ &\mathbf{D}_s = \left[\eta\mathbf{Y}_h\mathbf{A}_b^{s^{\mathbf{T}}} + \mu_2\left(\mathbf{G} + \frac{\mathbf{V}_3}{2\mu_2}\right)\mathbf{A}^{\mathbf{T}}\right]\left(\eta\mathbf{A}_b^s\mathbf{A}_b^{s^{\mathbf{T}}} + \mu_2\mathbf{A}\mathbf{A}^{\mathbf{T}}\right)^{-1}.\end{aligned} \quad (17)$$

From the above optimization for $\mathbf{D}_s$ and $\mathbf{A}^r$, it can be seen that spectral and spatial information provided by $\mathbf{Y}_h$ and $\mathbf{Y}_m$ can be effectively involved for the incompletely overlapped HM and LH image fusion, by exploiting the spectral similarity between the overlapping and nonoverlapping areas and the co-constraints applied to the representation error in both overlapping and nonoverlapping areas. By leveraging the closed-form solutions derived from (17), the optimized spectral dictionary can better represent the nonoverlapping area, thereby enabling more accurate reconstruction of the entire HH image. Finally, the proposed architecture and algorithm are shown in Fig. 2 and Algorithm 1, respectively. Besides, the penultimate line of Algorithm 1 denotes a reconstruction process, in which the reconstructed spectra in $\widehat{\mathbf{X}}^s$ and $\widehat{\mathbf{X}}^r$ are placed into the corresponding positions in the reconstructed HH images $\widehat{\mathbf{X}}$ via equation $\widehat{\mathbf{X}} = \widehat{\mathbf{X}}^s\mathbf{S}^{\mathbf{T}} + \widehat{\mathbf{X}}^r\mathbf{S}^{r\,\mathbf{T}}$.

In addition, two variants derived from (8) have also been developed to provide additional ways for comparisons. Specifically, for the first variant method, by only considering sparsity but not the low-rank constraint on $\mathbf{A}^r$ for the nonoverlapped HH image $\mathbf{X}^r$, the DEFs method is expressed as

$$\begin{aligned}\arg\min_{\mathbf{D}_s,\mathbf{A}^r}\ &\eta\left\|\mathbf{Y}_h - \mathbf{D}_s\mathbf{A}^s\mathbf{B}\right\|_F^2 + \left\|\mathbf{Y}_m^s - \mathbf{L}\mathbf{D}_s\mathbf{A}^s\right\|_F^2 \\ &+ \left\|\mathbf{Y}_m^r - \mathbf{L}\mathbf{D}_s\mathbf{A}^r\right\|_F^2 + \lambda\left\|\mathbf{A}^r\right\|_1.\end{aligned} \quad (18)$$

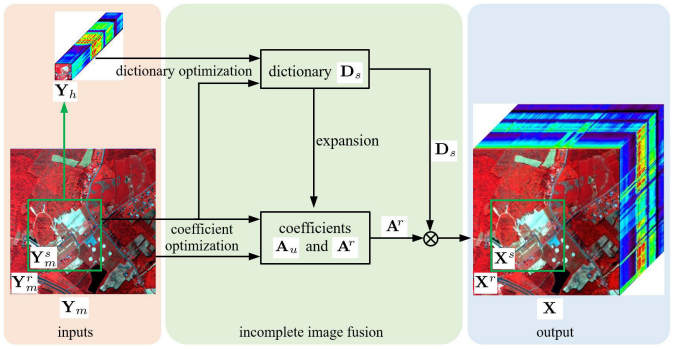


Fig. 2. Architecture of our proposed DEF method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Proposed DEF Algorithm

Input : $\mathbf{Y}_h$, $\mathbf{Y}_m$, $\mathbf{L}$, $\mathbf{B}$, K, ξ, η, γ, λ, μ_1, μ_2, T;
Compute $\widehat{\mathbf{X}}^s$ via same-scene-oriented fusion method;
For $k = 0, 1, \ldots, T-1$
  Compute $\mathbf{U}$ via Eq.(10);
  Compute $\mathbf{A}^r$ via Eq.(14);
  Compute $\mathbf{D}_s$ via Eq.(17);
End
$\widehat{\mathbf{X}}^r \leftarrow \mathbf{D}_s\mathbf{A}^r$;
$\widehat{\mathbf{X}} = \widehat{\mathbf{X}}^s\mathbf{S}^{\mathbf{T}} + \widehat{\mathbf{X}}^r\mathbf{S}^{r\mathbf{T}}$;
Output: $\widehat{\mathbf{X}}$ as the reconstructed $\mathbf{X}$.

We can see with (18) that the DEFs method has not considered the other characteristics in the nonoverlapping area, and this is the main difference between (8) and (18). For the second variant method, that is, the DEFr method, both sparse and low-rank constraints are applied to the nonoverlapped HH image $\mathbf{X}^r$. The low-rank property comes from the spectral similarity in $\mathbf{X}^r$, as under the representation of the spectral dictionary $\mathbf{D}s$, that is, $\mathbf{X}^r = \mathbf{D}_s\mathbf{A}^r + \mathbf{N}^r$, the corresponding coefficient matrix $\mathbf{A}^r$ has a low-rank characteristic. Thus, the optimization problem of the DEFr method includes both the sparse constraint expressed as $\|\mathbf{A}^r\|_1$ and the low-rank constraint expressed as $\gamma\|\mathbf{A}^r\|_*$

$$\arg\min_{\mathbf{D}_s,\mathbf{A}^r} \eta\left\|\mathbf{Y}_h - \mathbf{D}_s\mathbf{A}^s\mathbf{B}\right\|_F^2 + \left\|\mathbf{Y}_m^s - \mathbf{L}\mathbf{D}_s\mathbf{A}^s\right\|_F^2 + \left\|\mathbf{Y}_m^r - \mathbf{L}\mathbf{D}_s\mathbf{A}^r\right\|_F^2 + \lambda\left\|\mathbf{A}^r\right\|_1 + \gamma\left\|\mathbf{A}^r\right\|_*. \quad (19)$$

We can see with (19) that, the DEFr method has considered low-rank constraint on $\mathbf{A}^r$ in the nonoverlapped HH image $\mathbf{X}^r$, but has not considered the low-rank constraint on $[\mathbf{A}^r\ \mathbf{A}_u^s]$ in the entire HH image $\mathbf{X}$, which is the main difference between (8) and (19). Similar to the optimization process of (8), the closed-form solutions of (18) and (19) can also be calculated by using the ADMM algorithm. In summary, (8), (18), and (19) are three totally different optimization objectives with no equal relationship among them. And, we notate them as the DEF, DEFs, and DEFr methods just according to their different coverages: the entire HH image $\mathbf{X}$, the overlapped HH image $\mathbf{X}^s$ and the nonoverlapped HH image $\mathbf{X}^r$.

III. Experimental Results and Discussions

To compare the fusion results of the proposed DEF method, the unsupervised fusion spatial reconstruction network (UFS-Rnet) method [46] is used for comparison, and several same scene-oriented tangential fusion methods are also compared, which can directly apply the spectral dictionary or spectral mapping relationship established in the overlapping area to nonoverlapping area without any fine-tuning process, including the cluster-based fusion method using multibranch BPNNs (CF-BPNNs) [44], the unsupervised network-based fusion method (Unnet) [43] and the nonfactorization sparse representation and dictionary learning method (NFSRDL) [22]. Besides, two variants originally derived from (8) are also compared, including the DEFs and DEFr methods. Additionally, four full reference quality evaluation indices are utilized in the experiments, including the mean square error (MSE), the peak-signal-to-noise ratio (PSNR), the SAM, and the relative dimensionless global error in synthesis (ERGAS).

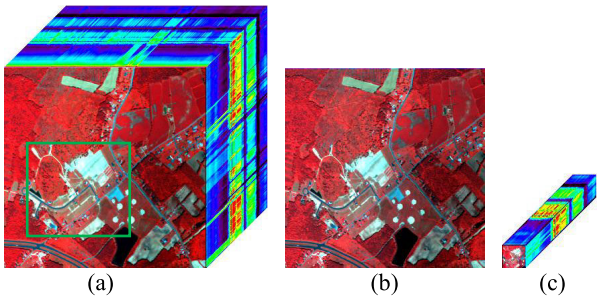


Fig. 3. False color images of the original and simulated AVIRIS images. (a) HH, (b) HM, and (c) LH.

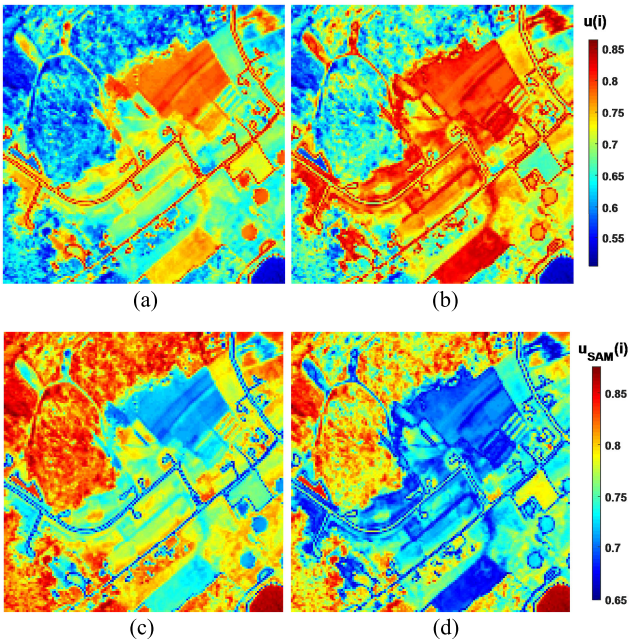


Fig. 4. Spectral maps measured by the similarity index $\boldsymbol{u}$ and SAM-scored similarity index $\boldsymbol{u}_{\mathrm{SAM}}$. (a) $\boldsymbol{u}$ in the hyperspectral domain. (b) $\boldsymbol{u}$ in the multispectral domain. (c) $\boldsymbol{u}_{\mathrm{SAM}}$ in the hyperspectral domain. (d) $\boldsymbol{u}_{\mathrm{SAM}}$ in the multispectral domain.

A. Experimental Results on the Simulated Dataset

The HH image collected by the AVIRIS airborne system in 1996 [52] with a dimension of 300 × 300 × 93 is used in this part of experiment, given in Fig. 3(a) as a false color image. For the overlapping area shown in the green box of Fig. 3(a), a subimage with a dimension of 150 × 150 × 93 is used as the overlapped HH image $\mathbf{X}^s$, and the remaining part is used as the nonoverlapped HH image $\mathbf{X}^r$. Utilizing the IKONOS-like spectral response function [53] as matrix $\mathbf{L}$, the HM image with a dimension of 300 × 300 × 4 is simulated as shown in Fig. 3(b). Since the spectral similarity between the overlapping and nonoverlapping areas is an important basis for the use of dictionary expansion and low-rank constraint in the DEF method, the spectral similarity maps measured by similarity index $\boldsymbol{u}$ are shown in the first row of Fig. 4, where Fig. 4(a) and (b) represents the spectral similarity in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

TABLE I
AVERAGED FUSION RESULTS ON THE AVIRIS DATASET

Method	the non-overlapping area				the entire area			
	MSE	PSNR	SAM	ERGAS	MSE	PSNR	SAM	ERGAS
UFSRnet [46]	0.4256	51.8411	1.2966	0.5776	0.4037	52.0707	1.2344	0.5196
CF-BPNNs [44]	0.3821	52.3085	1.2053	0.5270	0.3511	52.6767	1.1298	0.4688
Unnet [43]	0.4186	51.9127	1.2086	0.5575	0.3789	52.3454	1.1147	0.4900
NFSRDL [22]	0.5377	50.8256	1.3630	0.6303	0.4991	51.1490	1.2732	0.5522
DEFs	0.4679	51.4290	1.3071	0.5898	0.3777	52.3598	1.1373	0.4823
DEFr	0.4127	51.9743	1.2282	0.5442	0.3367	52.8587	1.0788	0.4507
DEF	**0.2998**	**53.3630**	**1.0688**	**0.4738**	**0.2528**	**54.1026**	**0.9604**	**0.3991**

hyperspectral domain and the spectrally degraded multispectral domain, respectively. The dark red corresponds to a large value of $\boldsymbol{u}$, indicating high spectral similarity. As can be seen from Fig. 4(a) that there is indeed a high spectral similarity between the overlapping and nonoverlapping areas, and the spectral similarity between the multispectral images in Fig. 4(b) can also reflect the spectral similarity in Fig. 4(a) to some extent. Furthermore, by adding arccosine calculation into (9) and replacing the minimum operator with maximum operator, the SAM index can be utilized to define a SAM-scored similarity index $\boldsymbol{u}_{\mathrm{SAM}}$, which can equivalently measure the spectral similarity from the view of one-to-many relationship. The corresponding spectral maps in the hyperspectral domain and the spectrally degraded multispectral domain are shown in the last row of Fig. 4. The dark blue corresponds to a small value of $\boldsymbol{u}_{\mathrm{SAM}}$, indicating high spectral similarity. The spectral maps measured by the SAM-scored similarity index $\boldsymbol{u}_{\mathrm{SAM}}$ shown in Fig. 4(c) and (d) are consistent with that measured by the similarity index $\boldsymbol{u}$ shown in Fig. 4(a) and (b), as an inversely proportional relationship. But the SAM-scored similarity index $\boldsymbol{u}_{\mathrm{SAM}}$ leads to an increase of computational complexity due to the additional arccosine calculation, which should not be ignored when calculating the many-to-many relationship between two large spectral sets in our proposed method. And, for the overlapping area, the corresponding LH image is simulated with a dimension of $30 \times 30 \times 93$, where $\mathbf{B}$ is constructed by a 5×5 Gaussian kernel and a 5-pixel down sampling rate in both directions. The LH image in false color is given as Fig. 3(c), of which the bands 11, 24, and 49 are used as inputs for three channels. It should also be emphasized that there is no corresponding LH image outside the green box of Fig. 3(a). Besides, our previous proposed optimized twin dictionary based fusion method [27] is used to reconstruct the overlapped HH image $\mathbf{X}^s$, and parameters K and ξ are set to 100 and 0.2; parameters η, γ, and λ are set to 1, 10^{-3}, and 10^{-4}, respectively. In addition, the default parameters in comparison methods are set as the corresponding original papers, to make a fair comparison.

Table I gives the incomplete fusion performances on the AVIRIS dataset in the nonoverlapping and entire areas with the averaged MSE, PSNR, SAM, and ERGAS indices. The bold and underlined represent the best and second-best results. As shown in Table I, our DEF method shows the best incomplete fusion performances than the other comparison methods in the nonoverlapping and the entire areas. Among all comparisons, compared with the deep learning-based UFSRnet, CF-BPNNs, Unnet methods, and the dictionary learning-based NFSRDL method, the MSE and SAM indices of the proposed DEF method are reduced by more than 0.08° and 0.14° in the nonoverlapping area, and 0.10° and 0.17° in the entire image, and the corresponding PSNR indices are improved by more than 1.05 and 1.42 dB. Compared with the DEFs method that only considers the sparsity and does not utilize the universal spectra, our proposed DEF method, which takes the spectral similarity of universal spectra into account, shows the best incomplete fusion results in the nonoverlapping and the entire area. Specifically, in the nonoverlapping area, our proposed DEF method remarkably improves the PSNR index by 1.93 dB and reduces the SAM index by 0.23°. In the entire area, our proposed DEF method improves fusion performance by 1.74 dB in the PSNR and 0.17° in SAM, compared with the DEF method only considering sparsity. Compared with the DEFr method that additionally considers the spectral similarity in the nonoverlapping area, our proposed DEF method takes the spectral similarity between the overlapping and nonoverlapping areas into account, which improves the PSNR index by 1.38 and 1.24 dB in the nonoverlapping and entire areas, respectively.

To show the visual comparison among different methods, the first row of Fig. 5 gives the reconstructed false-color HH images on the AVIRIS dataset, and the last two rows give the corresponding error images under the MSE and SAM indexes, where the dark blue and dark red represent the small and large errors, respectively. As can be seen from the false color images in Fig. 5, the reconstruction performance of our proposed method for the nonoverlapping area is comparable to that for the overlapping area, which reflects the effectiveness of the proposed DEF method in dealing with incompletely overlapped image fusion problem, especially in the nonoverlapping area without corresponding LH image. Furthermore, the error images in Fig. 5 indicate that by utilizing the spectral similarity constraints and the expanded spectral dictionary, our proposed DEF method gives better reconstruction results in the spatial and spectral domains.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

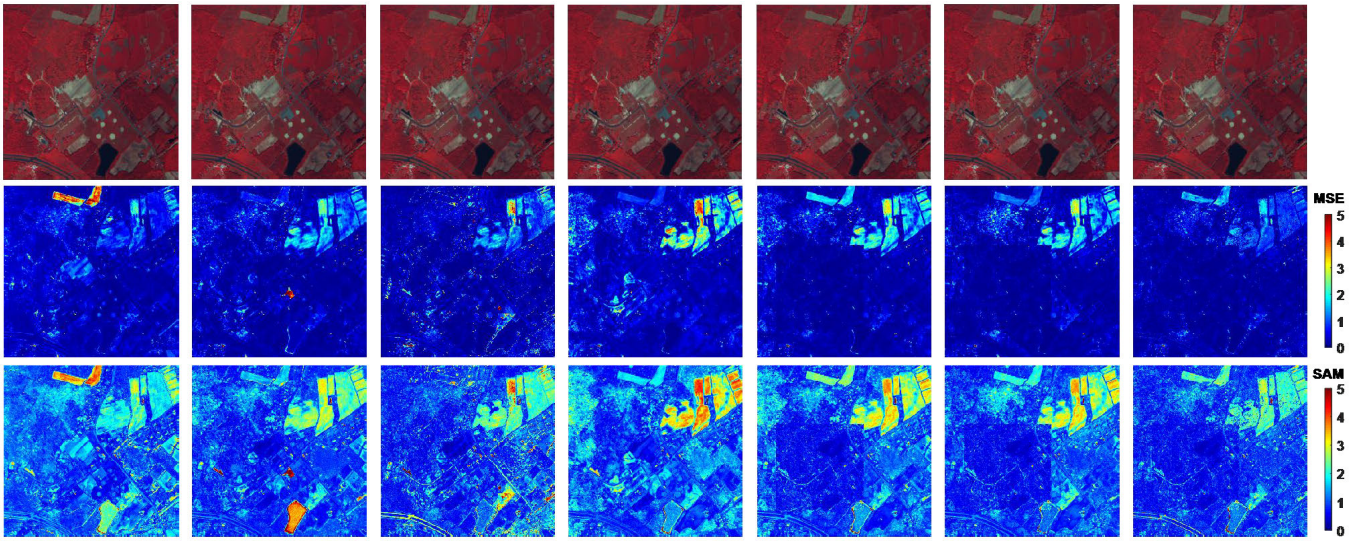


Fig. 5. False color and error images in MSE and SAM on the AVIRIS dataset. (a) UFSRnet, (b) CF-BPNNs, (c) Unnet, (d) NFSRDL, (e) DEFs, (f) DEFr, and (g) DEF.

TABLE II
AVERAGED FUSION RESULTS ON THE KSC DATASET

Method	the non-overlapping area				the entire area			
	MSE	PSNR	SAM	ERGAS	MSE	PSNR	SAM	ERGAS
UFSRnet [46]	0.2306	54.5025	0.5720	0.2243	0.1988	55.1466	0.5240	0.2097
CF-BPNNs [44]	0.1852	55.4551	0.5237	0.1949	0.1606	56.0744	0.4928	0.1822
Unnet [43]	0.2390	54.3477	0.5613	0.2248	0.2002	55.1164	0.5087	0.2065
NFSRDL [22]	0.2785	53.6830	0.6914	0.2701	0.2591	53.9961	0.6750	0.2598
DEFs	0.2406	54.3173	0.6078	0.2144	0.1933	55.2680	0.5364	0.1936
DEFr	0.1715	55.7891	0.5468	0.1942	0.1420	56.6085	0.4911	0.1767
DEF	**0.1395**	**56.6853**	**0.5138**	**0.1850**	**0.1171**	**57.4453**	**0.4632**	**0.1685**

B. Experimental Results on the KSC Dataset

To show the generalization of the DEF method to different scenes, another hyperspectral dataset with a dimension of 300 × 300 × 93 captured by AVIRIS airborne system over the Kennedy Space Center (KSC) [54] is also used as the HH image in this part of the experiment. The overlapping area is located just in the center of the entire area, with a dimension of 150 × 150 × 93.

Utilizing the same parameters as in Section III-A, Table II shows the incomplete fusion performances in the entire and nonoverlapping areas evaluated by the four full-reference quality evaluations. In the nonoverlapping area, the proposed DEF method outperforms in spatial and spectral reconstruction, with at least 0.03° and 0.01° reduced in MSE and SAM indices. Compared with the CF-BPNNs, Unnet and NFSRDL methods that directly apply the spectral mapping or spectral dictionary established in the low-spatial-resolution overlapping area to the high-spatial-resolution nonoverlapping area, our proposed DEF method expands the spectral dictionary of the overlapping area to the nonoverlapping area, and its optimization takes both areas into account, resulting in an improvement of 1.23 dB in the PSNR and a reduction of 0.05 in MSE. Compared to the DEF method that only considers the sparsity of $\mathbf{A}^r$, and the DEFr method that considers the sparsity and low-rank constraints of $\mathbf{A}^r$, our proposed DEF method, utilizing the spectral similarity between the overlapping and nonoverlapping areas, improves the PSNR index by 2.36 and 0.89 dB and reduces the SAM index by 0.09° and 0.03°, respectively. In the entire area, compared with the DEFs and DEFr methods, our proposed method improves 2.17 and 0.83 dB in the PSNR and reduces 0.07° and 0.02° in the SAM index. The improvement of the PSNR and SAM indices in the nonoverlapping area is even better than that of the entire area. It indicates the superiority and generality of our DEF method to deal with incompletely overlapped multispectral and hyperspectral image fusion.

Additionally, the corresponding visual results are given in Fig. 6, where the first row gives the reconstructed false-color HH images on the KSC dataset, and the last two rows are the MSE and SAM error maps, respectively. The comparable fusion results between the overlapping and nonoverlapping areas shown in the first row of Fig. 6(g) indicate the effectiveness of the proposed DEF method in dealing with the incompletely overlapped image fusion. Moreover, the darkest

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

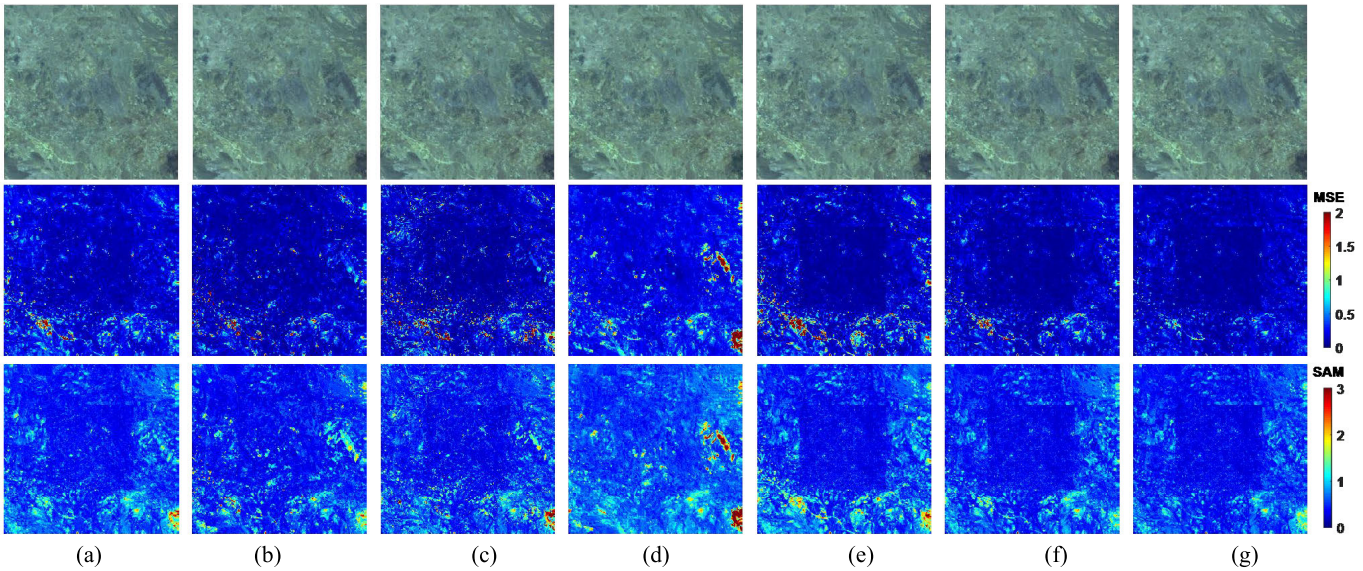


Fig. 6. False color and error images in MSE and SAM on the KSC dataset. (a) UFSRnet, (b) CF-BPNNs, (c) Unnet, (d) NFSRDL, (e) DEFs, (f) DEFr, and (g) DEF.

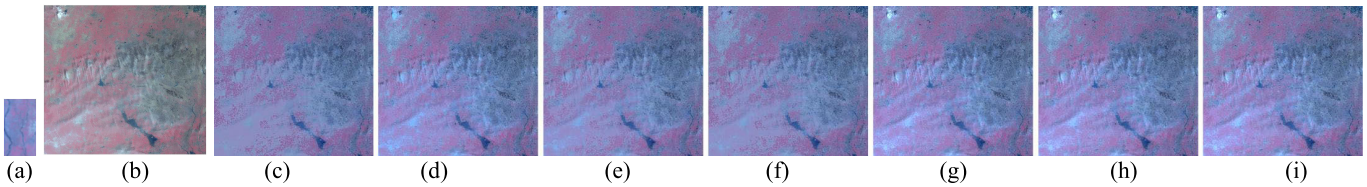


Fig. 7. False color images of the original and fused images with different methods on the real HJ-1A dataset. (a) LH, (b) HM, (c) UFSRnet, (d) CF-BPNNs, (e) Unnet, (f) NFSRDL, (g) DEFs, (h) DEFr, and (i) DEF.

blue maps in the last two rows of Fig. 6(g) show not only the best fusion performance of our proposed method in terms of spectral and spatial reconstruction, but also the applicability of the proposed method to incompletely overlapped image fusion on different scenes.

C. Experimental Results on the Real Dataset

To evaluate the incompletely overlapped image fusion performance of our proposed DEF method in real applications, the real hyperspectral and multispectral images captured by the HJ-1A satellite in Shandong Province of China on March 23, 2019 [55], are utilized in this part of the experiment. The image captured by the hyperspectral camera in the overlapping area with a dimension of $589 \times 350 \times 3$, and the image captured by the CCD camera with a dimension of $11800 \times 12540 \times 4$ are used as the LH and HM images, respectively. Besides, their corresponding false color images are shown in Fig. 7(a) and (b), in which bands 80, 55, and 18 of the LH image and bands 4, 2, and 1 of the HM image are used as the inputs for three channels.

Since there is no ground-truth HH image for quantitative comparison in this real dataset, Fig. 7 only gives the false color images of the enlarged LH by 7 times in both directions, the original HM and the fused HH images with different methods for visual comparison. Compared with the LH image shown in Fig. 7(a), our DEF method preserves most of the spectral information, and the spectral distortion is smaller compared with the blue biased fusion result of the CF-BPNNs method in Fig. 7(d). Compared with the HM image shown in Fig. 7(b), our proposed method reconstructs most of the texture information in the spatial domain, even for such a small overlapping area, which is nearly 80 times less than the entire nonoverlapped HM image. The above incompletely overlapped fusion performance shows the superiority of the DEF method for real applications.

D. Discussions on Parameter Settings

Considering that the parameters n, η, γ, and λ affect the incomplete fusion performance through spectral similarity, low-rank, and sparsity, we have modified these parameters and the signal-to-noise ratio (SNR) on the AVIRIS dataset to perform a sensitivity analysis of our proposed method. Besides, the fusion performances on different overlapping areas are also compared. Furthermore, the impact of individual components, such as our proposed dictionary expansion, universal spectra, and optimization process, on the fusion performance is also evaluated in ablation experiments.

The number of universal spectra n is an important parameter that affects the quality of the final fused result. The number of universal spectra n is related to the similarity ratio ξ, which is expressed as $n = M\xi$. From this point of view, Fig. 8(a) shows the relations between fusion performance on the varied parameter n. The PSNR curve shows that our proposed DEF method performs stably when the number of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

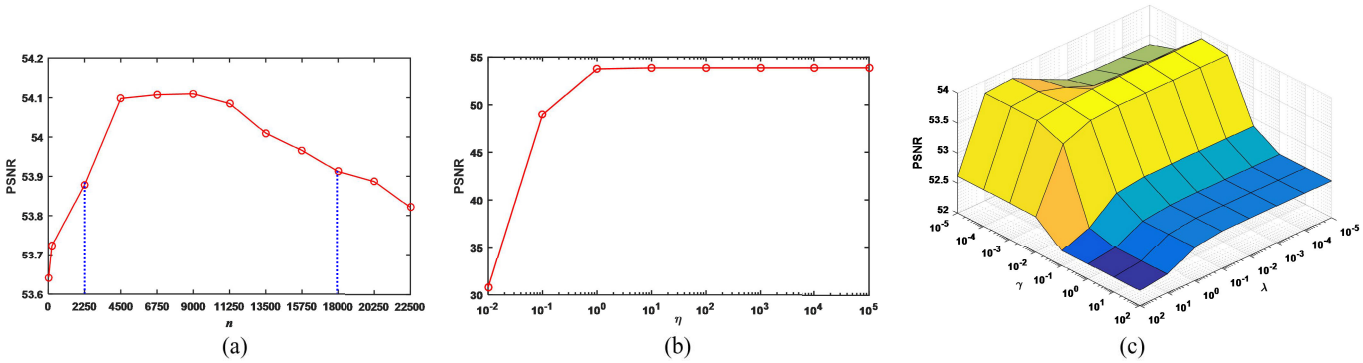


Fig. 8. Relations between the PSNR and parameters. (a) n, (b) η, and (c) γ and λ on the AVIRIS dataset.

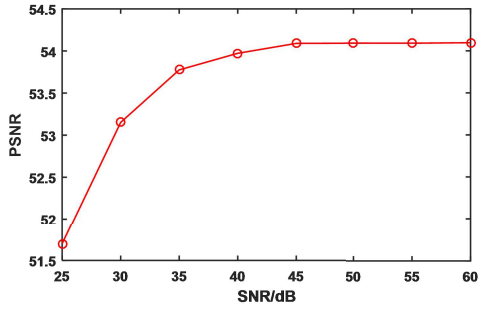


Fig. 9. PSNR curve with varied SNRs on the AVIRIS dataset.

universal spectra n is in the range of [2250, 18000]. However, as the number of universal spectra increases, the degree of similarity between the universal spectra $\mathbf{X}_u^s$ and $\mathbf{X}^r$ may decrease, which affects the low-rank characteristics and thus affects fusion performance to some extent. Fig. 8(b) gives the incomplete fusion performances on the varied parameter η, in which our method shows better performance when η is larger than 1. And, Fig. 7(c) gives the relation between the PSNR and parameters λ and γ. The proposed incompletely overlapped image fusion method performs better when λ is less than 1 and γ is in the range of $[10^{-3}, 10^{-2}]$. Besides, Fig. 9 gives the fusion results on varied SNRs. As shown in Fig. 9, the proposed method without any denoising process performs stably with the SNR higher than 40 dB.

In addition, since different overlapping areas would cover different ground objects, Table III gives the fusion results for different overlapping areas evaluated by the four full-reference quality indices, and the corresponding visualization results are given in Fig. 10. In particular, the first row gives the reconstructed false color images, where the green boxes denote the different overlapping areas. The last two rows give the fusion results evaluated by error images on the MSE and SAM indices. The above fusion results show that, although different overlapping areas can affect the fusion performance to some extent, the fusion results of our DEF method are very similar.

To further evaluate the effectiveness of our proposed dictionary expansion, universal spectra, and optimization process, on the fusion performance, the fusion results for possible combinations of the above components are given in Table IV, where the bold represents the best result. Among them, the

TABLE III

FUSION PERFORMANCES ON THE AVIRIS DATASET WITH DIFFERENT OVERLAPPING AREAS

areas	MSE	PSNR	SAM	ERGAS
(a)	0.3652	52.5058	1.1824	0.4987
(b)	0.2581	54.0121	0.9999	0.3974
(c)	0.3065	53.2666	1.0789	0.4398

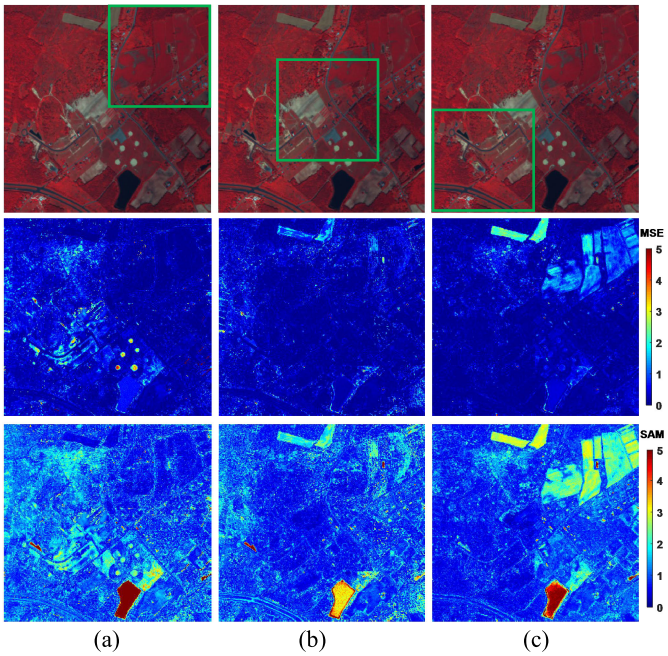


Fig. 10. False color and error images in MSE and SAM on the AVIRIS dataset with different overlapping areas. (a) Upper right. (b) Center. (c) Lower left.

first row of Table IV statistically evaluated the fusion performance without using dictionary expansion and universal spectra (termed as DEF-ds), whose optimization process has no low-rank constraints. The second row of Table IV shows the fusion performance only using universal spectra without dictionary expansion (termed as DEF-d), whose optimization process comes with sparse and low-rank constraints. In the above DEF-ds and DEF-d methods without dictionary expansion, the spectral dictionary $\mathbf{D}_s$ is obtained directly from the overlapped LH image $\mathbf{Y}_h$ without applying (15). The last row

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
AVERAGED FUSION RESULTS WITH DIFFERENT COMPONENTS OF OUR PROPOSED METHOD ON THE AVIRIS DATASET

	possible combinations				evaluation results							
	dictionary expansion	universal spectra	optimization process		the non-overlapping area				the entire area			
			sparse	low-rank	MSE	PSNR	SAM	ERGAS	MSE	PSNR	SAM	ERGAS
DEF-ds	×	×	√	×	0.4762	51.3533	1.3170	0.5767	0.3834	52.2946	1.1427	0.4726
DEF-d	×	√	√	√	0.4243	51.8545	1.2426	0.5565	0.3452	52.7495	1.0894	0.4587
DEF	√	√	√	√	**0.2998**	**53.3630**	**1.0688**	**0.4738**	**0.2528**	**54.1026**	**0.9604**	**0.3991**

of Table IV gives the fusion performance of our proposed DEF method, which utilizes universal spectra and applies our proposed dictionary expansion via (15). Compared with the DEF-ds method without using dictionary expansion and universal spectra, the DEF-d method, which takes the spectral similarity of universal spectra and low-rank constraints into account, improves the PSNR index by 0.50 dB and reduces the SAM index by 0.07° in the nonoverlapping area, and improves the PSNR index by 0.45 dB and reduces the SAM index by 0.05° in the entire area. Compared with the DEF-d method that directly applies the spectral dictionary learned in the overlapping area to the nonoverlapping area, our proposed DEF method expands the spectral dictionary learned over the overlapping area to the nonoverlapping area and optimizes it by taking both areas into account under the sparse and low-rank constraints, resulting in an improvement of 1.51 dB in the PSNR and a reduction of 0.17° in SAM in the nonoverlapping area, and an improvement of 1.35 dB in the PSNR and a reduction of 0.12° in SAM in the entire area. The above ablation experiments show the effectiveness of our proposed dictionary expansion, the universal spectra, and the optimization process.

E. Computational Complexity Analysis

The computational complexity of the DEF method depends on calculating the expanded dictionary and the sparse coefficient. Among them, computational complexity of optimizing the sparse coefficient $\mathbf{A}^r$ and its splitting variables $\mathbf{E}$ and $\mathbf{F}$ shown in (14) are $O\left(\lambda_y K^2+\lambda_y K(N-M)+K^2(N-M)+K^3\right)$, $O(N-M)$, and $O(K^2(N-M+n)+K(N-M+n)^2)$. And that of calculating spectral dictionary $\mathbf{D}_s$ and the splitting variable $\mathbf{G}$ shown in (17) are $O(\lambda_x KN+\lambda_x K^2+K^2N+K^3)$ and $O(\lambda_x KN+\lambda_x^2\lambda_y+\lambda_x^2N+\lambda_x^3)$, respectively.

IV. CONCLUSION

This article proposed a new dictionary expansion scheme and a dictionary expansion-based multi- and hyperspectral image fusion method under the incompletely overlapping conditions, which can reconstruct the entire HH image, including both the overlapping and the nonoverlapping areas. Specifically, the incomplete fusion problem is divided into one fusion problem in the overlapping area and one reconstruction problem over the nonoverlapping area separately, where the former is formulated as a traditional spectral dictionary learning-based fusion process, and the latter is formulated as a new dictionary expansion-based reconstruction process only using the HM image. By leveraging the new dictionary expansion scheme, the spectral dictionary learned in the overlapping area is expanded and fine-tuned to fit with the nonoverlapping area, with some specific universal spectra that have high spectral similarity crossing both areas. Furthermore, the sparse coefficient matrix in the nonoverlapping area is optimized, with a spectral similarity constraint over the overlapping and the nonoverlapping areas. Experimental results on three different datasets with different coverage areas indicated that our proposed DEF method performs better than the other related or tangentially related state-of-the-art methods for the incomplete fusion problem. Furthermore, experimental results on the HJ-1A satellite dataset also demonstrated the superiority of our proposed DEF method in practical applications.

REFERENCES

[1] C. -H. Lin and S. -S. Young, "Signal subspace identification for incomplete hyperspectral image with applications to various inverse problems," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512316.
[2] Y. Noboru, Y. Ozasa, and M. Tanaka, "Hyperspectral image dataset for individual penguin identification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Athens, Greece, Jul. 2024, pp. 9383–9387.
[3] A. Medellin, A. Bhamri, R. Langari, and S. Gopalswamy, "Real-time semantic segmentation using hyperspectral images for unstructured and unknown environments," in *Proc. 13th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Athens, Greece, Oct. 2023, pp. 1–5.
[4] S. P. Ang, S. L. Phung, L. Bui, and A. Bouzerdoum, "AdaptorNAS: A new perturbation-based neural architecture search for hyperspectral image segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 3, pp. 1559–1571, Mar. 2024.
[5] H. Yang, H. Yu, K. Zheng, J. Hu, T. Tao, and Q. Zhang, "Hyperspectral image classification based on interactive transformer and CNN with multilevel feature fusion network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[6] X. Han, Y. Wei, H. Zhang, Q. Xu, and W. Sun, "Tuning to real for single-spectrum hyperspectral target detection," in *Proc. 13th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Athens, Greece, Oct. 2023, pp. 1–5.
[7] C. I. Chang, C. Y. Lin, and P. F. Hu, "Band sampling of hyperspectral anomaly detection in effective anomaly space," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502729.
[8] G. Vivone, "Multispectral and hyperspectral image fusion in remote sensing: A survey," *Inf. Fusion*, vol. 89, pp. 405–417, Jan. 2023.
[9] D. Sara, A. K. Mandava, A. Kumar, S. Duela, and A. Jude, "Hyperspectral and multispectral image fusion techniques for high resolution applications: A review," *Earth Sci. Informat.*, vol. 14, no. 4, pp. 1685–1705, Dec. 2021.
[10] Z. Chen, H. Pu, B. Wang, and G.-M. Jiang, "Fusion of hyperspectral and multispectral images: A novel framework based on generalization of pan-sharpening methods," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 8, pp. 1418–1422, Aug. 2014.
[11] M. Selva, B. Aiazzi, F. Butera, L. Chiarantini, and S. Baronti, "Hyper-sharpening: A first approach on SIM-GA data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3008–3024, Jun. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.

[12] C. Grohnfeldt, X. X. Zhu, and R. Bamler, "Jointly sparse fusion of hyperspectral and multispectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2013, pp. 4090–4093.

[13] J. Liang, Y. Zhang, and S. Mei, "Hyperspectral and multispectral image fusion using dual-source localized dictionary pair," in *Proc. Int. Symp. Intell. Signal Process. Commun. Syst. (ISPACS)*, Xiamen, China, Nov. 2017, pp. 261–264.

[14] T. Zhao, Y. Zhang, X. Xue, and M. He, "Hyperspectral and multispectral image fusion using collaborative representation with local adaptive dictionary pair," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 7212–7215.

[15] K. Arias, E. Vargas, and H. Arguello, "Hyperspectral and multispectral image fusion based on a non-locally centralized sparse model and adaptive spatial–spectral dictionaries," in *Proc. 27th Eur. Signal Process. Conf. (EUSIPCO)*, A Coruna, Spain, Sep. 2019, pp. 1–5.

[16] L. Sui, L. Li, J. Li, N. Chen, and Y. Jiao, "Fusion of hyperspectral and multispectral images based on a Bayesian nonparametric approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1205–1218, Apr. 2019.

[17] E. Ghasrodashti, A. Karami, R. Heylen, and P. Scheunders, "Spatial resolution enhancement of hyperspectral images using spectral unmixing and Bayesian sparse representation," *Remote Sens.*, vol. 9, no. 6, p. 541, May 2017.

[18] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[19] N. Akhtar, F. Shafait, and A. Mian, "Sparse spatio-spectral representation for hyperspectral image super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2014, pp. 63–78.

[20] X. Han, J. Luo, J. Yu, and W. Sun, "Hyperspectral image fusion based on non-factorization sparse representation and error matrix estimation," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2017, pp. 1155–1159.

[21] W. Dong et al., "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.

[22] X. Han, J. Yu, and W. Sun, "Hyperspectral image super-resolution based on non-factorization sparse representation and dictionary learning," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Beijing, China, Sep. 2017, pp. 963–966.

[23] X. Fu, S. Jia, M. Xu, J. Zhou, and Q. Li, "Sparsity constrained fusion of hyperspectral and multispectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[24] F. Zhang and K. Zhang, "Superpixel guided structure sparsity for multispectral and hyperspectral image fusion over couple dictionary," *Multimedia Tools Appl.*, vol. 79, nos. 7–8, pp. 4949–4964, Feb. 2020.

[25] J. Zhang, J. Lu, C. Wang, and S. Li, "Hyperspectral and multispectral image fusion via superpixel-based weighted nuclear norm minimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521612.

[26] Y. Zhang, L. Zhang, R. Song, C. Huang, and Q. Tong, "Considering nonoverlapped bands construction: A general dictionary learning framework for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505215.

[27] X. Han, J. Yu, J.-H. Xue, and W. Sun, "Hyperspectral and multispectral image fusion using optimized twin dictionaries," *IEEE Trans. Image Process.*, vol. 29, pp. 4709–4720, 2020.

[28] J. Xue, Y.-Q. Zhao, Y. Bu, W. Liao, J. C. Chan, and W. Philips, "Spatial–spectral structured sparse low-rank representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 3084–3097, 2021.

[29] C. Yi, Y.-Q. Zhao, and J. C. Chan, "Hyperspectral image super-resolution based on spatial and spectral correlation fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4165–4177, Jul. 2018.

[30] C. I. Kanatsoulis, X. Fu, N. D. Sidiropoulos, and W.-K. Ma, "Hyperspectral super-resolution: A coupled tensor factorization approach," *IEEE Trans. Signal Process.*, vol. 66, no. 24, pp. 6503–6517, Dec. 2018.

[31] J. Zeng, G. Zhou, Y. Qiu, Y. Ma, and Q. Zhao, "Hyperspectral and multispectral image fusion via Bayesian nonlocal CP factorization," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[32] R. Dian, L. Fang, and S. Li, "Hyperspectral image super-resolution via non-local sparse tensor factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3862–3871.

[33] W. He, Y. Chen, N. Yokoya, C. Li, and Q. Zhao, "Hyperspectral super-resolution via coupled tensor ring factorization," *Pattern Recognit.*, vol. 122, Feb. 2022, Art. no. 108280.

[34] J. Zhang, L. Zhu, C. Deng, and S. Li, "Hyperspectral and multispectral image fusion via logarithmic low-rank tensor ring decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11583–11597, 2024.

[35] T. Xu et al., "A coupled tensor double-factor method for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5515417.

[36] F. Zhou, R. Hang, Q. Liu, and X. Yuan, "Pyramid fully convolutional network for hyperspectral and multispectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 5, pp. 1549–1558, May 2019.

[37] L. Zhang, J. Nie, W. Wei, Y. Li, and Y. Zhang, "Deep blind hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2388–2400, Jun. 2021.

[38] M. Peng, L. Zhang, X. Sun, and Y. Cen, "Improving spectral resolution of multispectral data using convolutional neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Yokohama, Japan, Jul. 2019, pp. 3145–3148.

[39] J.-F. Hu, T.-Z. Huang, L.-J. Deng, H.-X. Dou, D. Hong, and G. Vivone, "Fusformer: A transformer-based fusion network for hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[40] L. Sun et al., "MDC-FusFormer: Multiscale deep cross-fusion transformer network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528914.

[41] R. Li, L. Zhang, Z. Wang, and X. Li, "MIMFormer: Multiscale inception mixer transformer for hyperspectral and multispectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 15122–15135, 2024.

[42] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Reciprocal transformer for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102148.

[43] J. Li, K. Zheng, L. Ni, and L. Gao, "A new unsupervised network for hyperspectral and multispectral image fusion," in *Proc. 13th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Athens, Greece, Oct. 2023, pp. 1–5.

[44] X. Han, J. Yu, J. Luo, and W. Sun, "Hyperspectral and multispectral image fusion using cluster-based multi-branch BP neural networks," *Remote Sens.*, vol. 11, no. 10, p. 1173, May 2019.

[45] S. Liu, S. Miao, S. Liu, B. Li, W. Hu, and Y.-D. Zhang, "Circle-net: An unsupervised lightweight-attention cyclic network for hyperspectral and multispectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4499–4515, 2023.

[46] H. Yu et al., "Unsupervised deep adaptive learning spatial reconstruction network based on hyperspectral data fusion," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Athens, Greece, Jul. 2024, pp. 9366–9369.

[47] Q. Wang, C. Wu, Q. Li, and J. Li, "Chinese HJ-1A/B satellites and data characteristics," *Sci. China Earth Sci.*, vol. 53, no. S1, pp. 51–57, Dec. 2010.

[48] *CBERS-1D*. Accessed: Sep. 10, 2024. [Online]. Available: http://114.116.226.59/chinese/satellite/chinese/zy102d

[49] F. A. Kruse et al., "The spectral image processing system (SIPS)—Interactive visualization and analysis of imaging spectrometer data," *Remote Sens. Environ.*, vol. 44, nos. 2–3, pp. 145–163, 1993.

[50] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2011.

[51] G. H. Golub and C. Reinsch, "Singular value decomposition and least squares solutions," *Numerische Math.*, vol. 14, no. 5, pp. 403–420, Apr. 1970.

[52] *AVIRIS Datset*. Accessed: Sep. 10, 2017. [Online]. Available: http://aviris.jpl.nasa.gov

[53] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

[54] *Kennedy Space Center (KSC) Datset*. Accessed: Sep. 10, 2017. [Online]. Available: http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[55] *HJ-1A Dataset*. Accessed: Apr. 1, 2019. [Online]. Available: http://218.247.138.119:7777/DSSPlatform/index.html

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.



Xiaolin Han (Member, IEEE) received the B.S. degree in electronic engineering from Beijing Institute of Technology, Beijing, China, in 2015, and the Ph.D. degree in electronic engineering from Tsinghua University, Beijing, in 2020.

From September 2018 to March 2019, she was a Visiting Student with the Department of Statistical Science, University College London, London, U.K. From 2020 to 2022, she was a Post-Doctoral Researcher with Tsinghua University. She is currently an Associate Professor with the School of Mechatronical Engineering, Beijing Institute of Technology. Her research interests include hyperspectral image fusion, spectral super-resolution, and their applications.



Jiaxun Li received the B.S. degree from the School of Mechatronical Engineering, Beijing Institute of Technology, Beijing, China, in 2024, where she is currently pursuing the M.S. degree.

Her research interests include hyperspectral image fusion and applications.



Wei Wang (Member, IEEE) received the B.Eng., M.Sc., and Ph.D. degrees in science and technology of aeronautical and astronautical from Beijing Institute of Technology, Beijing, China, in 2007, 2009, and 2017, respectively.

He is currently an Associate Professor with the School of Aerospace Engineering, Beijing Institute of Technology. He has authored more than 100 peer-reviewed articles. His research interests include flight vehicle guidance and control.

Dr. Wang serves as a reviewer for many international journals such as IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS, IEEE TRANSACTIONS ON AEROSPACE AND ELECTRONIC SYSTEMS, *Journal of the Franklin Institute*, and *Chinese Journal of Aeronautics*.



Lijuan Niu received the bachelor's degree from Shandong Medical University, Shandong, China, in 1989.

She is currently a Chief Physician with the Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China. Her research interest covers artificial intelligence, multimodal-based image processing, and computer-aided diagnosis.



Weidong Sun (Member, IEEE) received the Ph.D. degree in information engineering from the University of Tokyo, Tokyo, Japan, in 1988.

He is currently a Professor with the Department of Electronic Engineering, Tsinghua University, Beijing, China. His research interests include image processing, pattern recognition, spatial information processing, and their applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:54:55 UTC from IEEE Xplore. Restrictions apply.